

L5

Assignment 1

- For both PSY305/508
- Assignment 1: Deadline: 11th September
- Introduction
- Review
- Conclusion on the basis of review and future application
- 500 words: Strict deadline
- bibliography (not included in word limit)
- figures
- Plag check with open source
- 2 days to resubmit if plag > 15%
- Please read the instruction document carefully

RUBRIC

Ideal Word Distribution (Excluding Citations):

Introduction ~ 200 words

Review ~ 200 words

Conclusion ~ 100 words

1. Introduction:

- Clear definition of the topic
- Figure if necessary to express the idea (the figure should be cited)
- Technical terms explained

2. Review:

- Highlight the studies.
- The studies should address an aspect of the topic.
- They should not be a compilation of studies that have keywords.

3. Conclusion

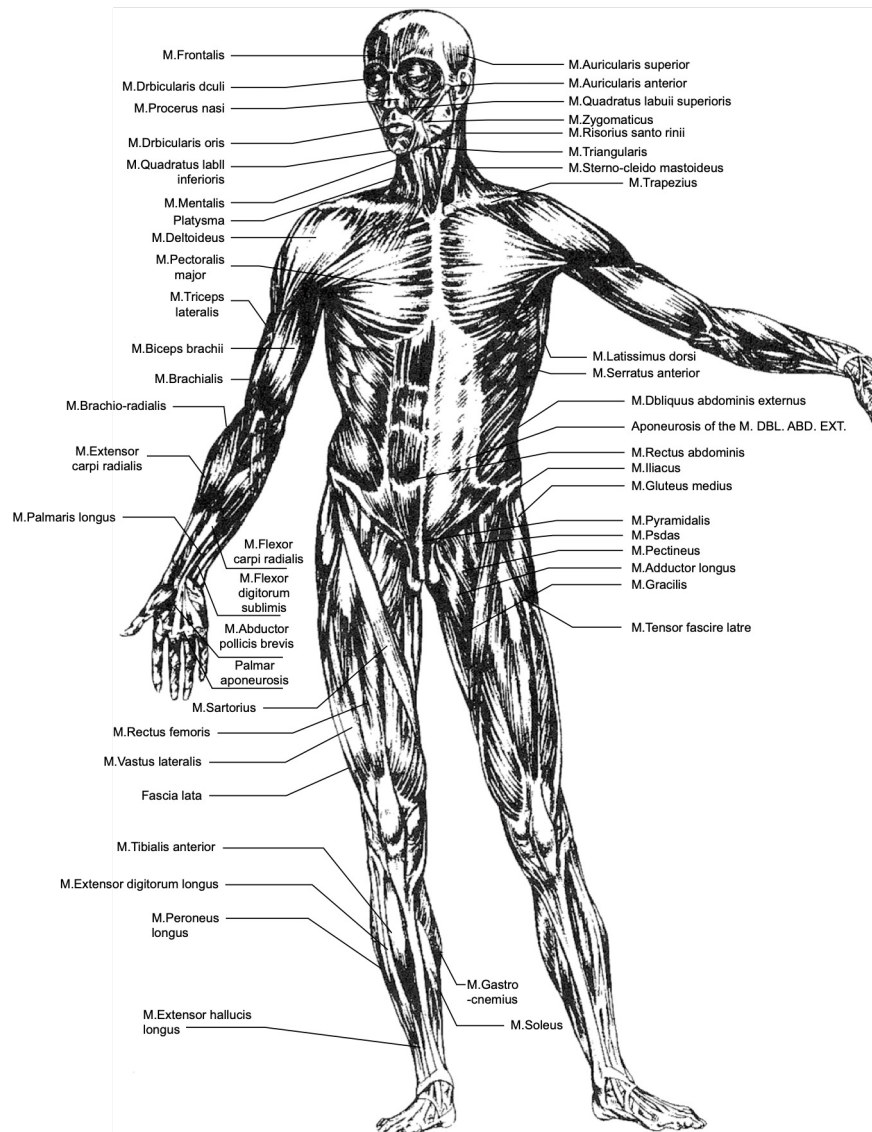
- Definition of the topic and one/two sentences of your conclusion as to what aspects it addresses of the topic and how?
- Abstract discussion will be marked average → 3 marks
- How well you have described the implications of the topic in societal development. Maybe your ideas no matter how unachievable they are.

Motor Movement

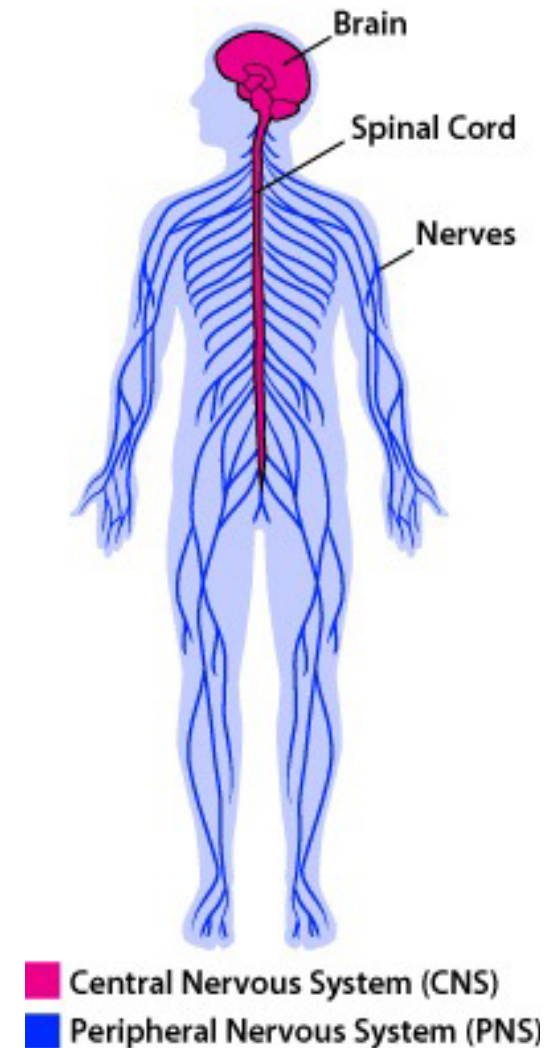
1. The *degrees of freedom*: How are movements selected out of infinite options to achieve particular tasks?
2. The *sequencing and timing* : How are behaviors ordered in time?
3. The *perceptual-motor integration* : How are perception and motor control combined?
4. The *learning* : How are perceptual-motor skills acquired?

Nervous and Muscular system

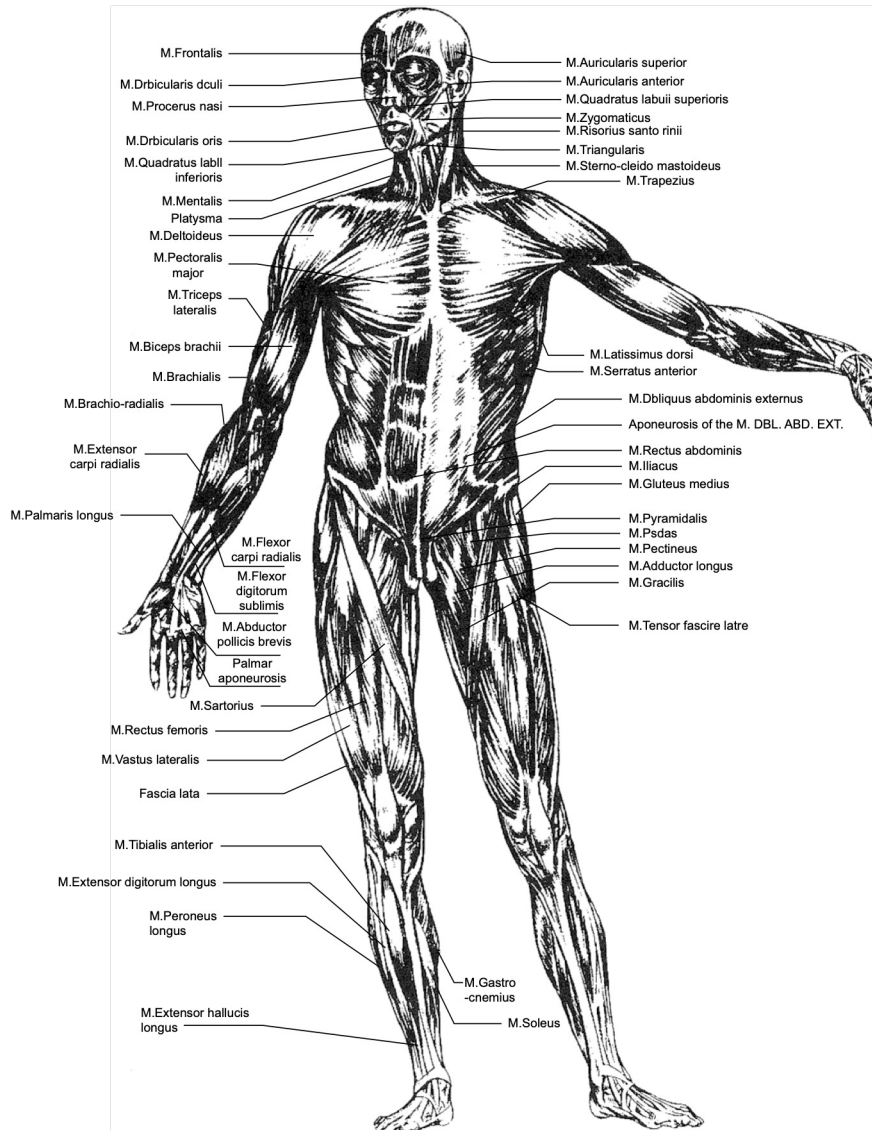
Musculature



Nervous system



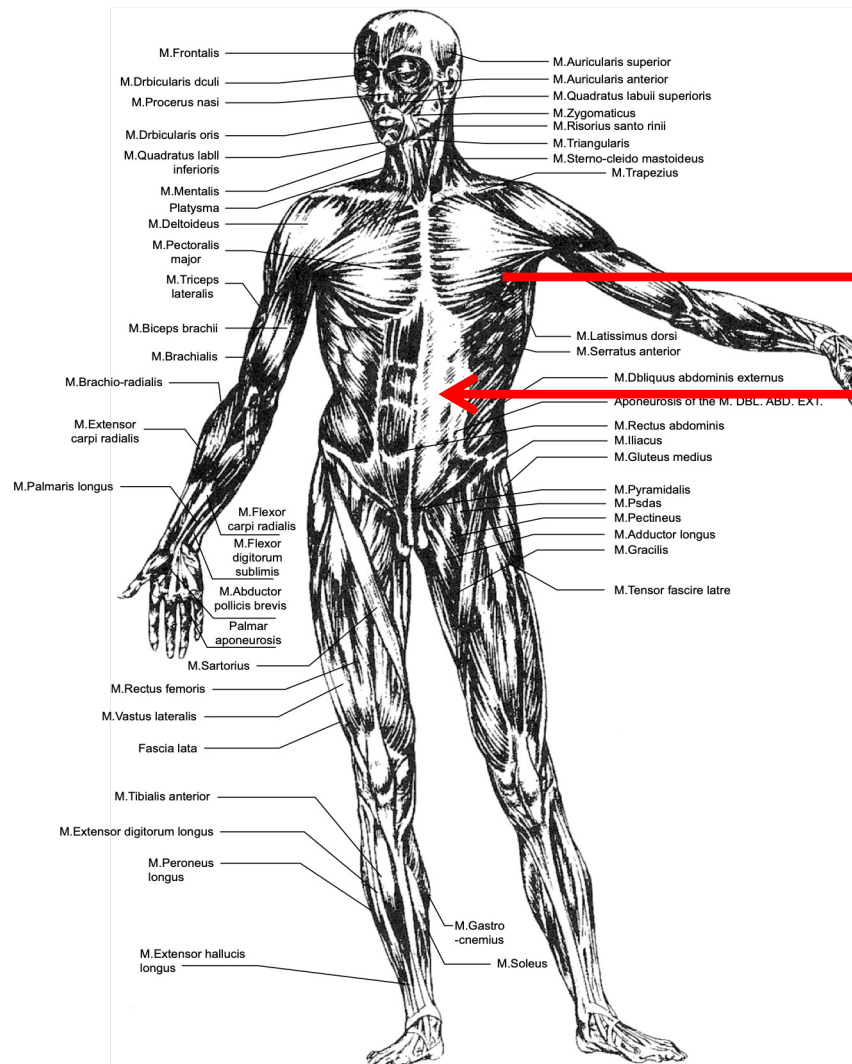
Muscles



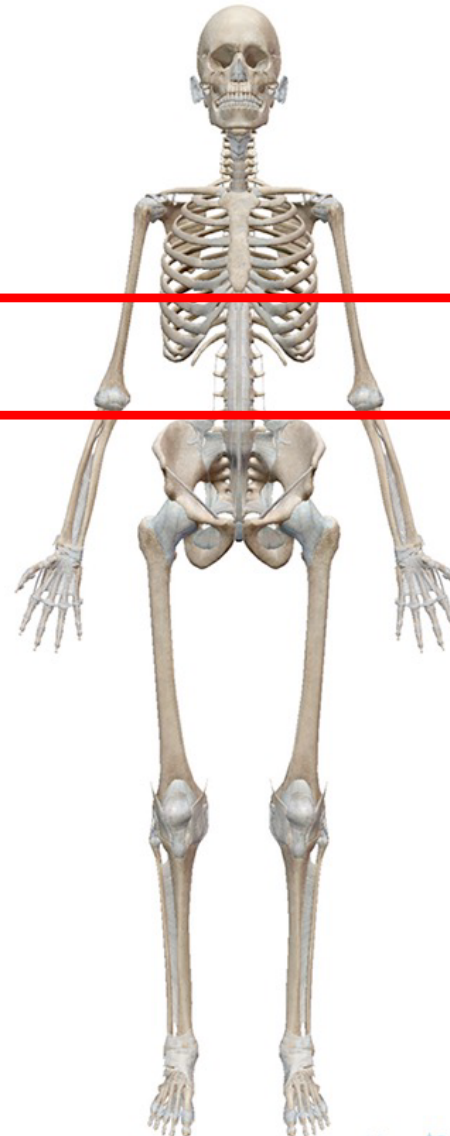
- Drives movement by contraction
- Three types:
 - **Skeletal:** Voluntary control
 - **Smooth:** Involuntary control (body organs)
 - **Cardiac muscles:** Involuntary rhythmic actions.

Nervous and Muscular system

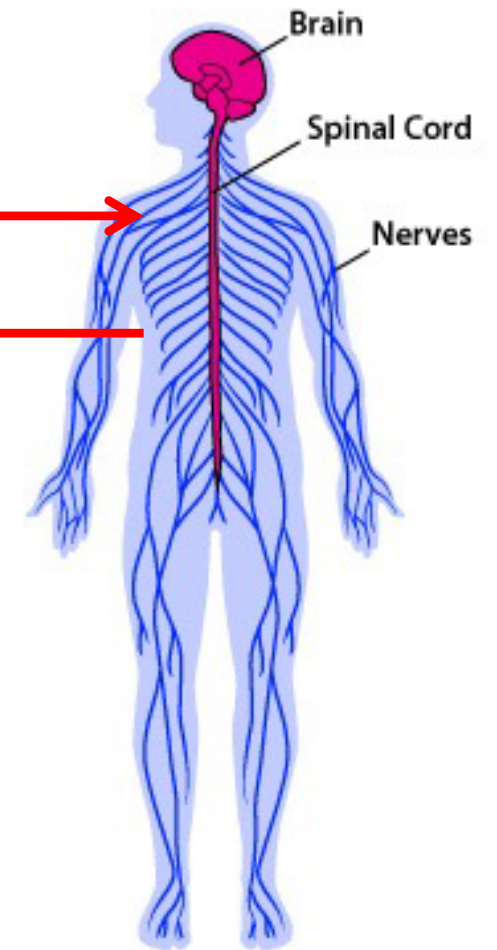
Musculature



Skeletal system

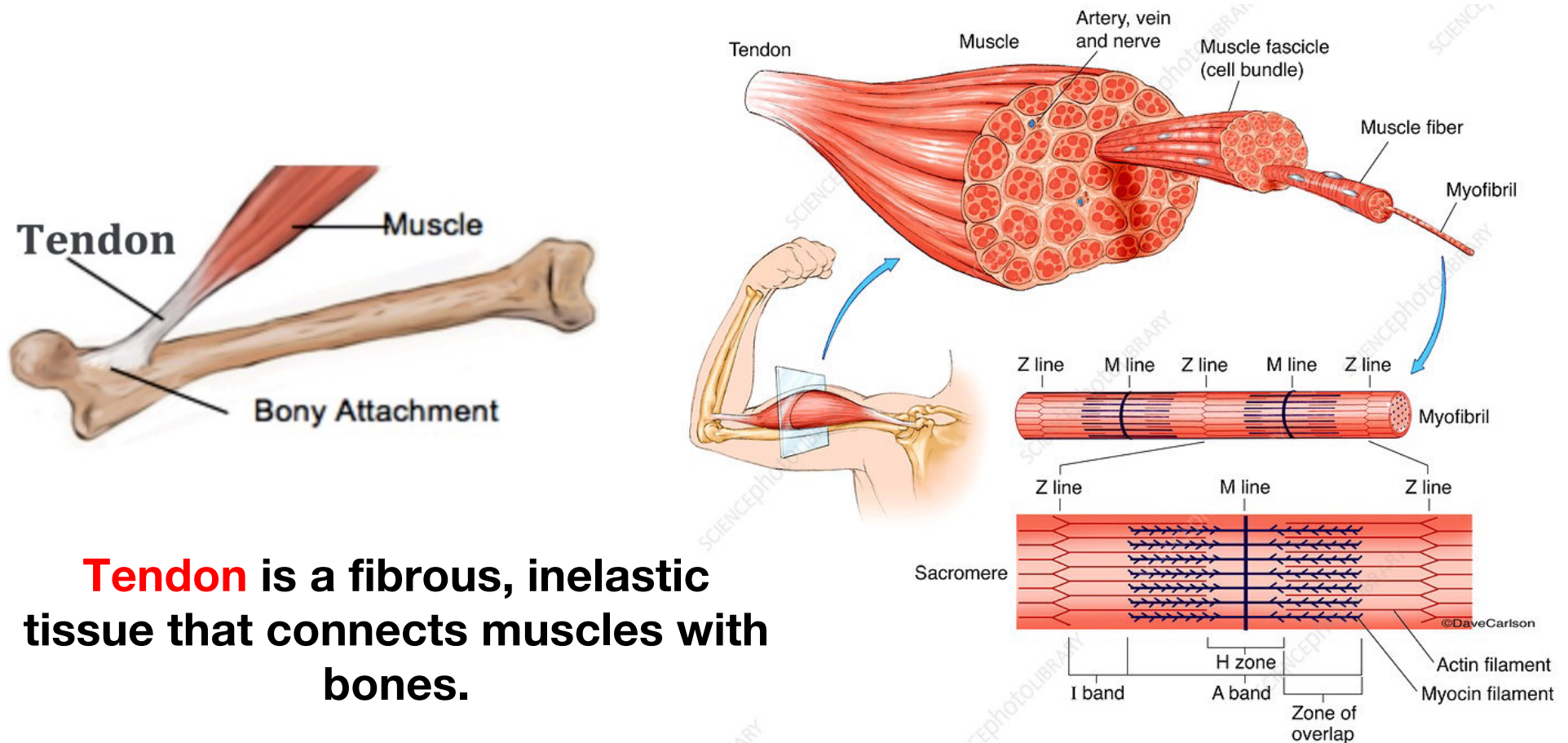


Nervous system



■ Central Nervous System (CNS)
■ Peripheral Nervous System (PNS)

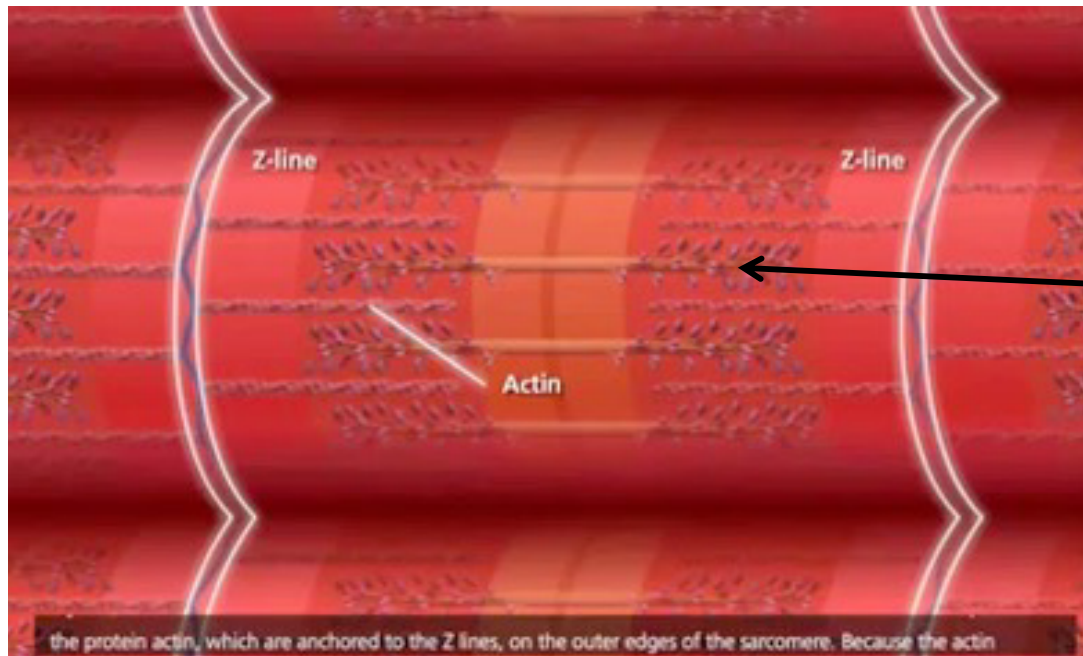
Skeletal muscle structure



Tendon is a fibrous, inelastic tissue that connects muscles with bones.

Sarcomere is a unit of skeletal muscle

Skeletal muscle in action

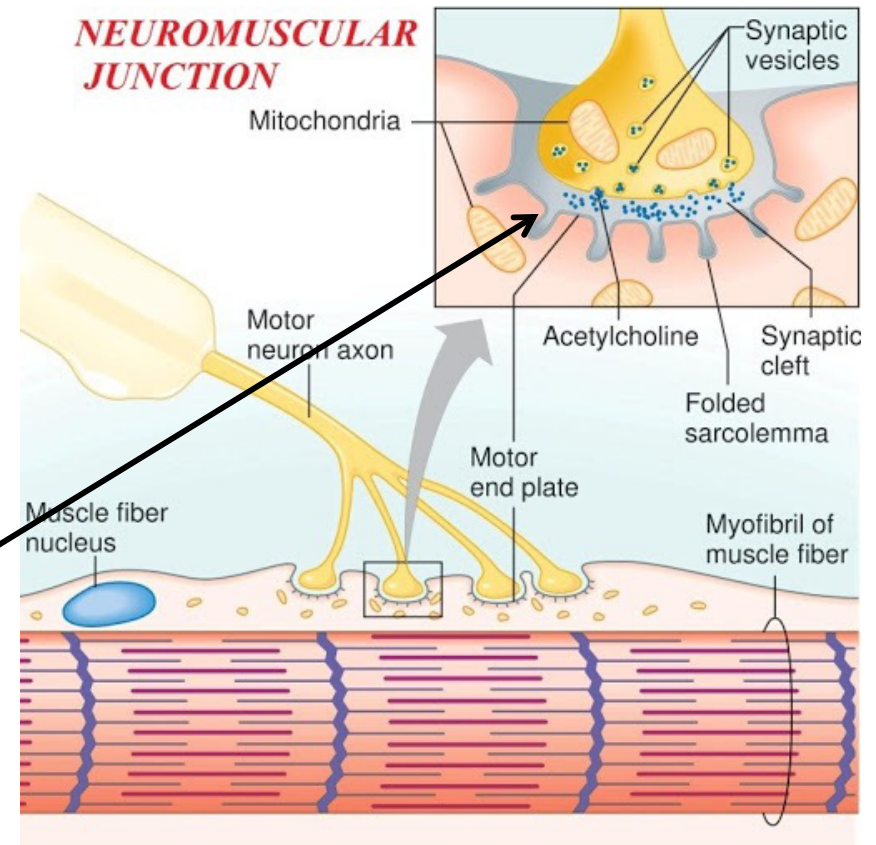
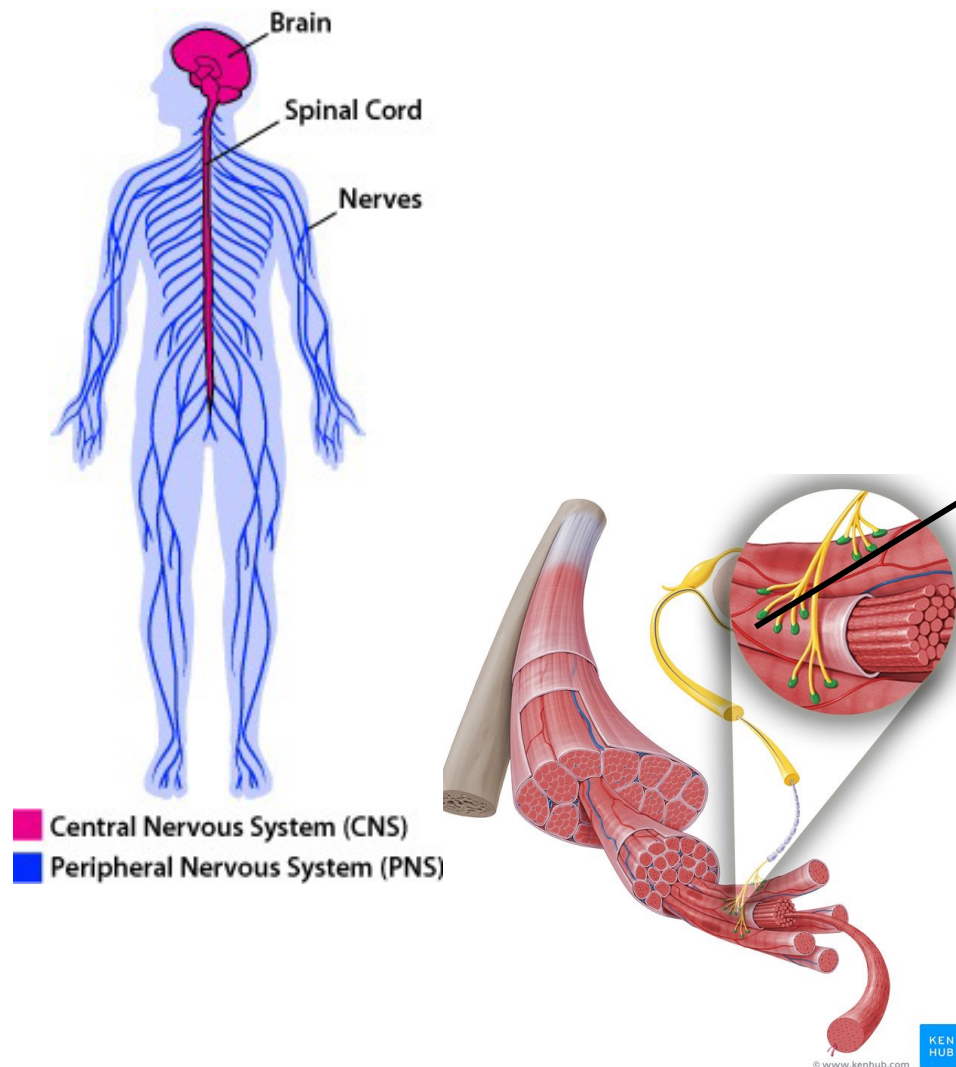


Myosin

How does information reaches to muscles to contract?

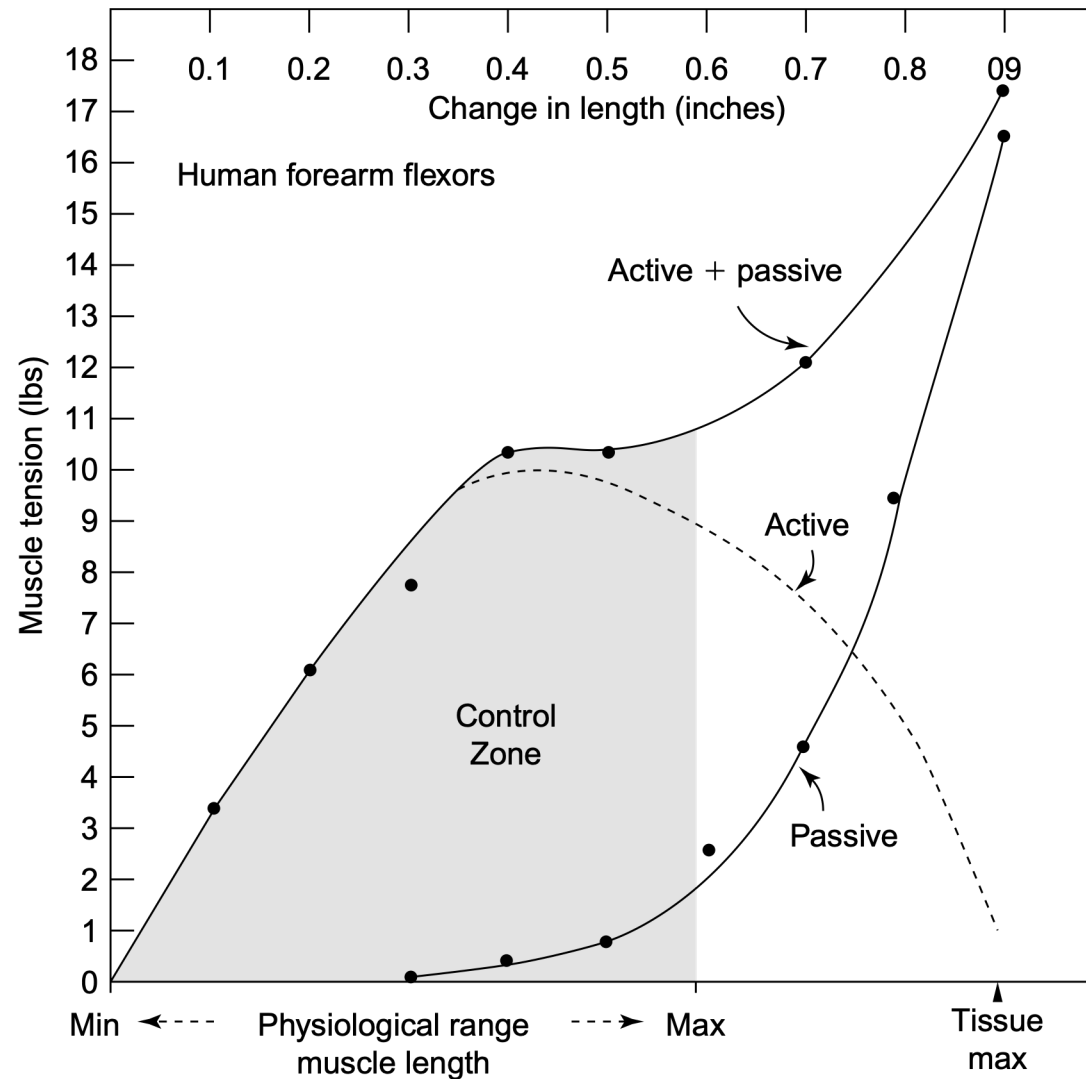
Neuromuscular junction

Nervous system

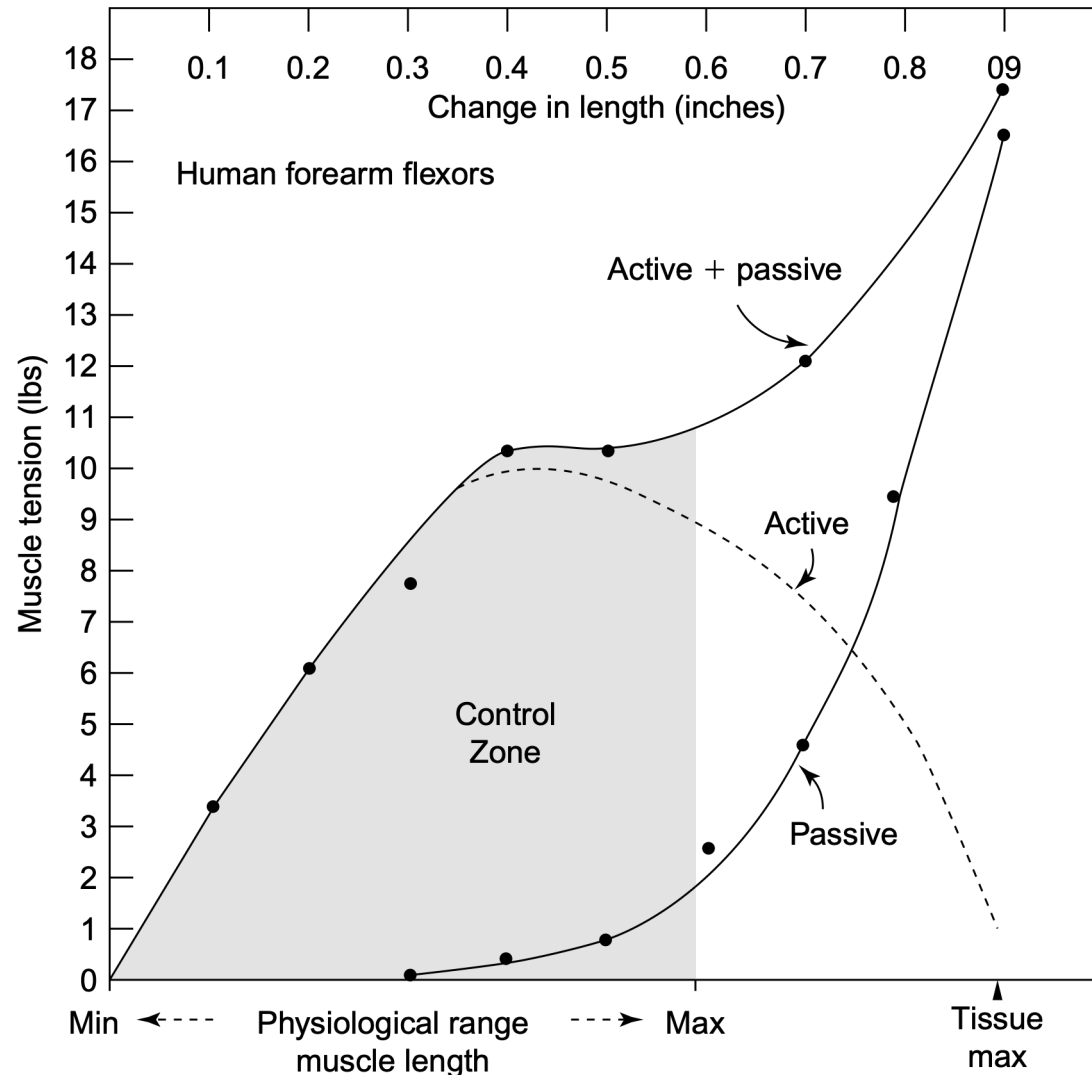


Acetylcholine is the neurotransmitter that helps in transmitting information from motor neurons to muscles to contract.

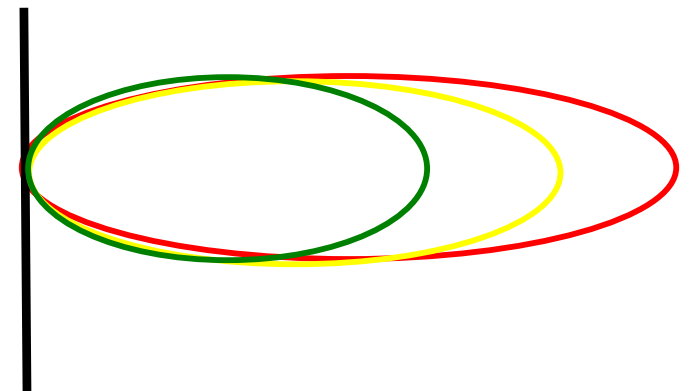
Muscle: Length-Tension relation



Passive tension: Recoil

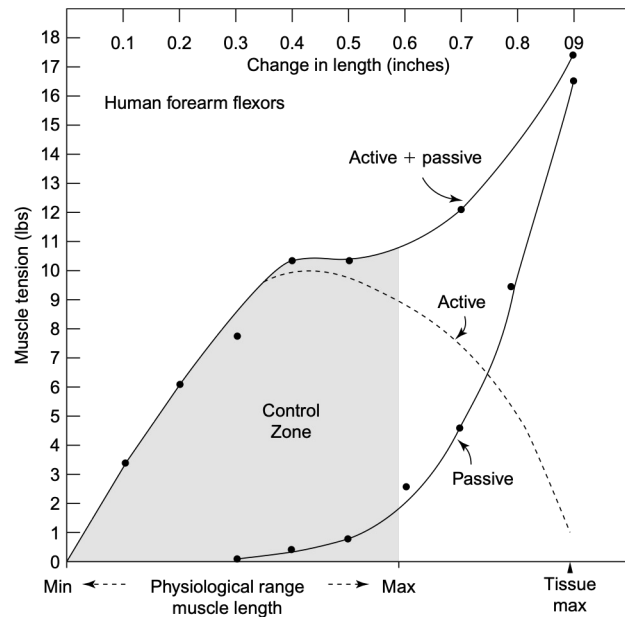


<https://depositphotos.com/vector-images/slshot-cartoon.html>



No Energy

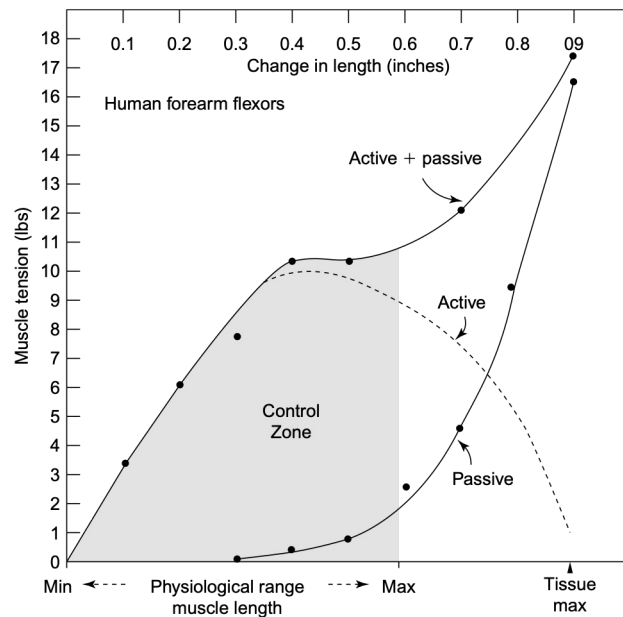
Passive tension: Recoil tension



Creates Tension in muscle to recoil

No Energy

Active tension: Contraction

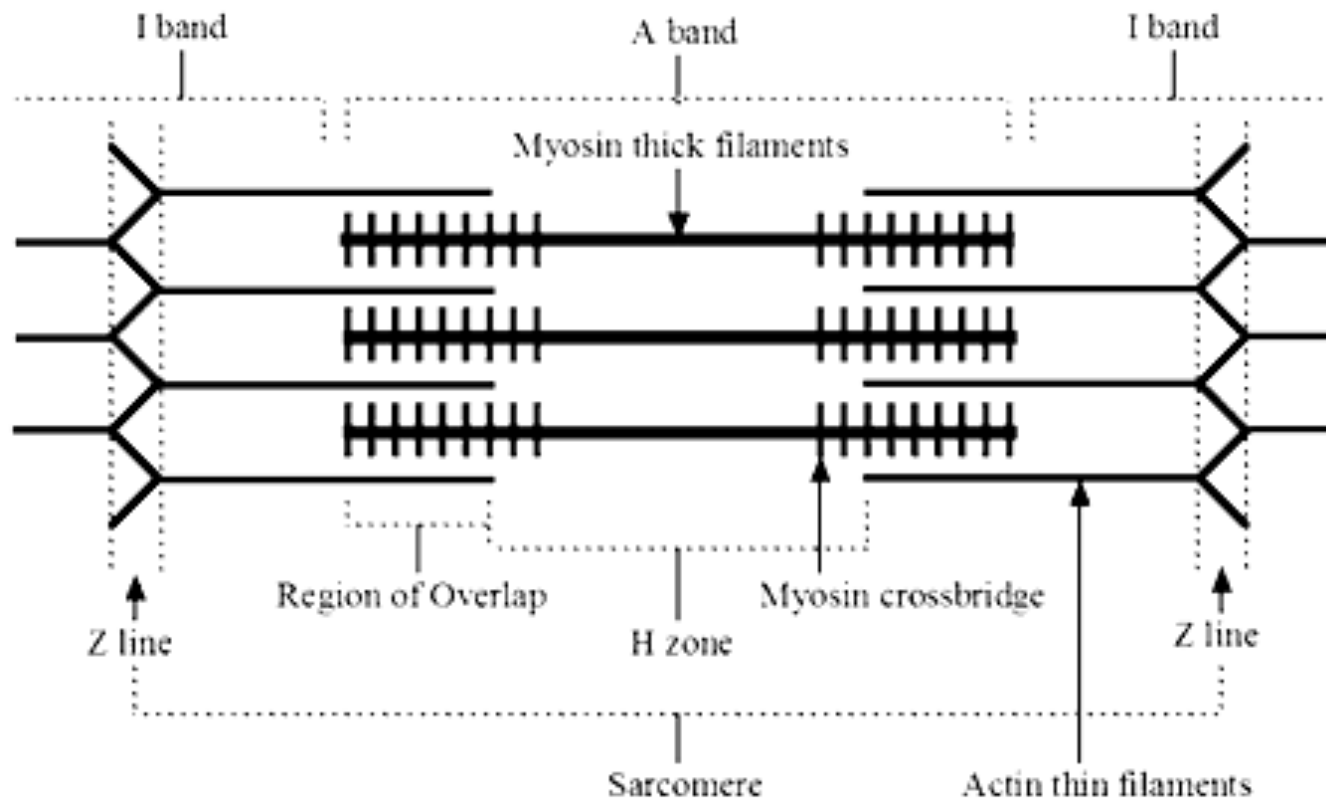


Actively contracts the muscle

Requires acetylcholine

Energy required

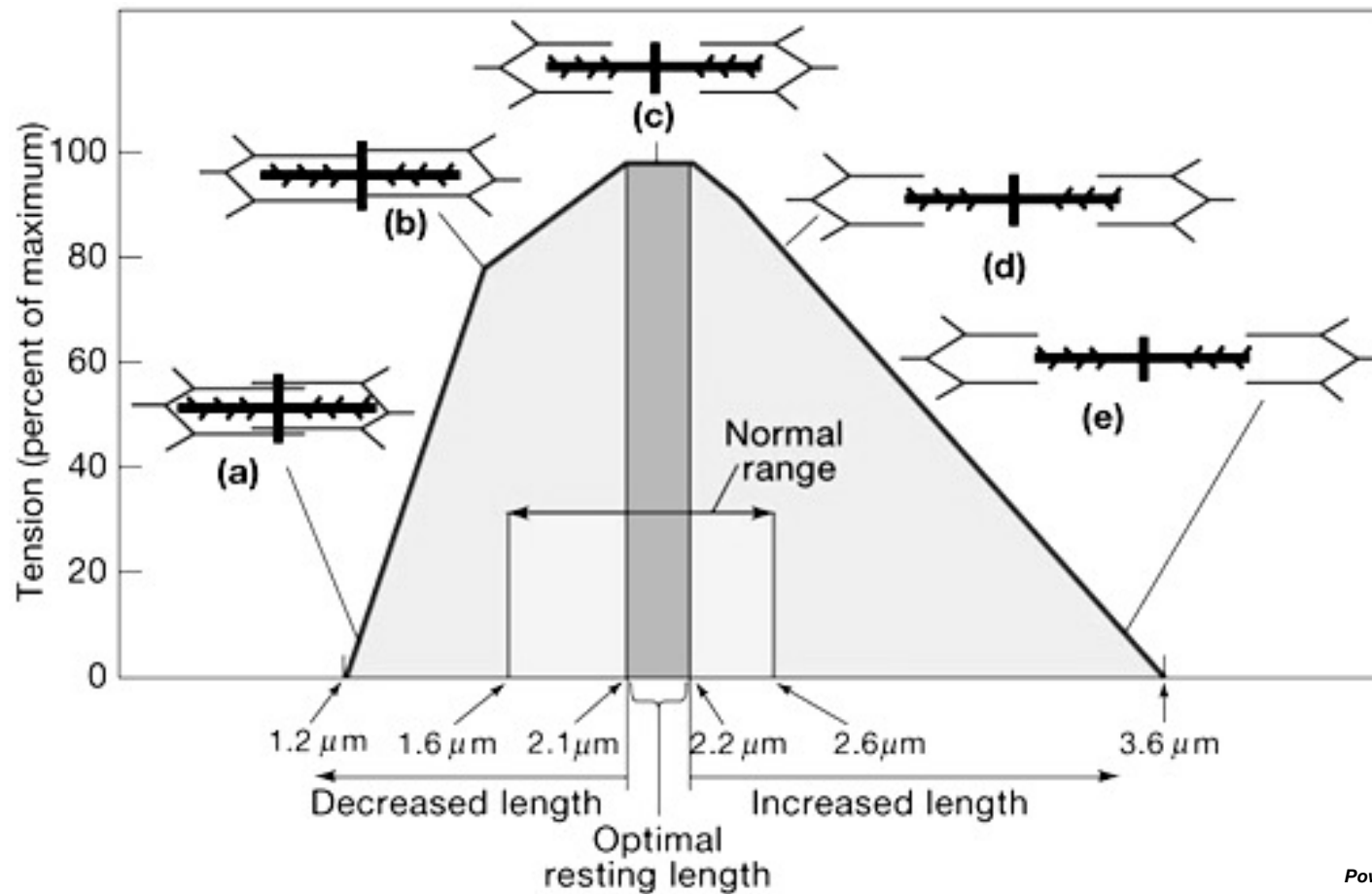
Active tension: Contraction



Power, 2012

Energy required

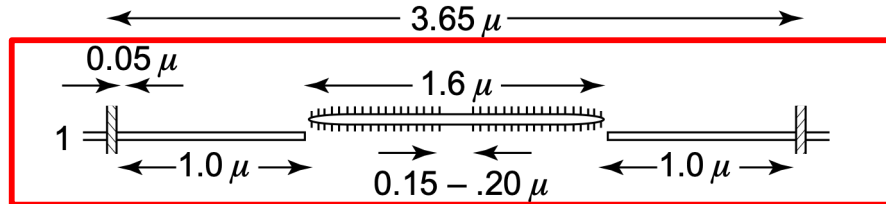
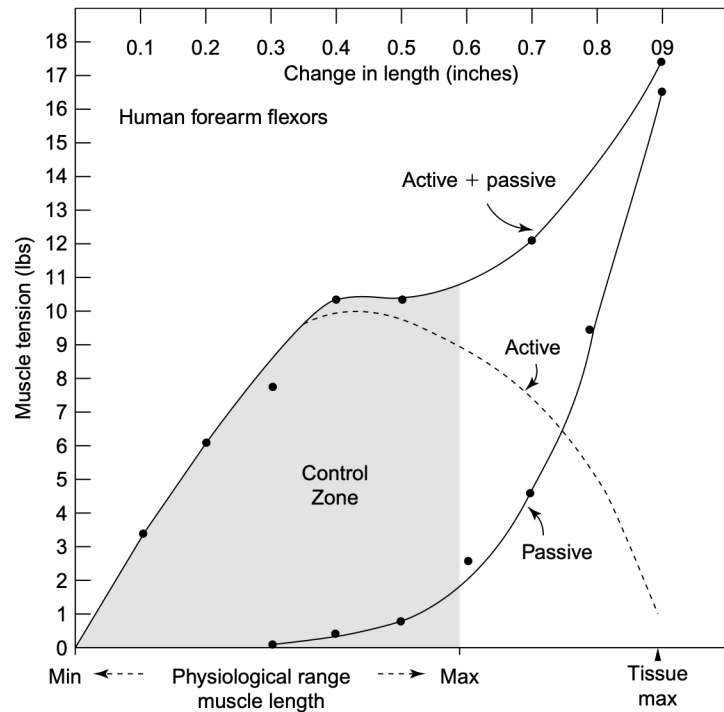
Active tension: Contraction



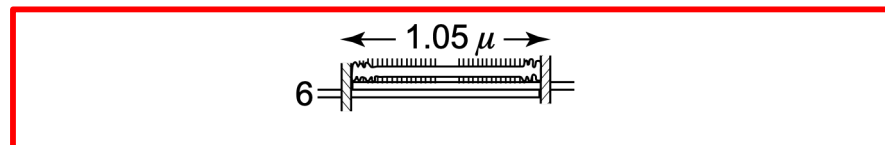
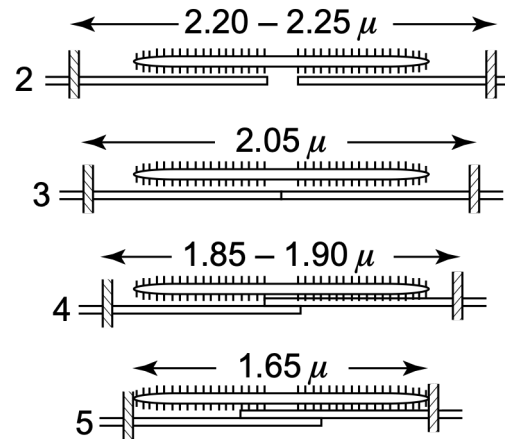
Power, 2012

- Max active force generated around resting stage of the muscle.
- Muscle had narrow range over which tension does not vary with the length (usual contraction)

Active tension: Contraction



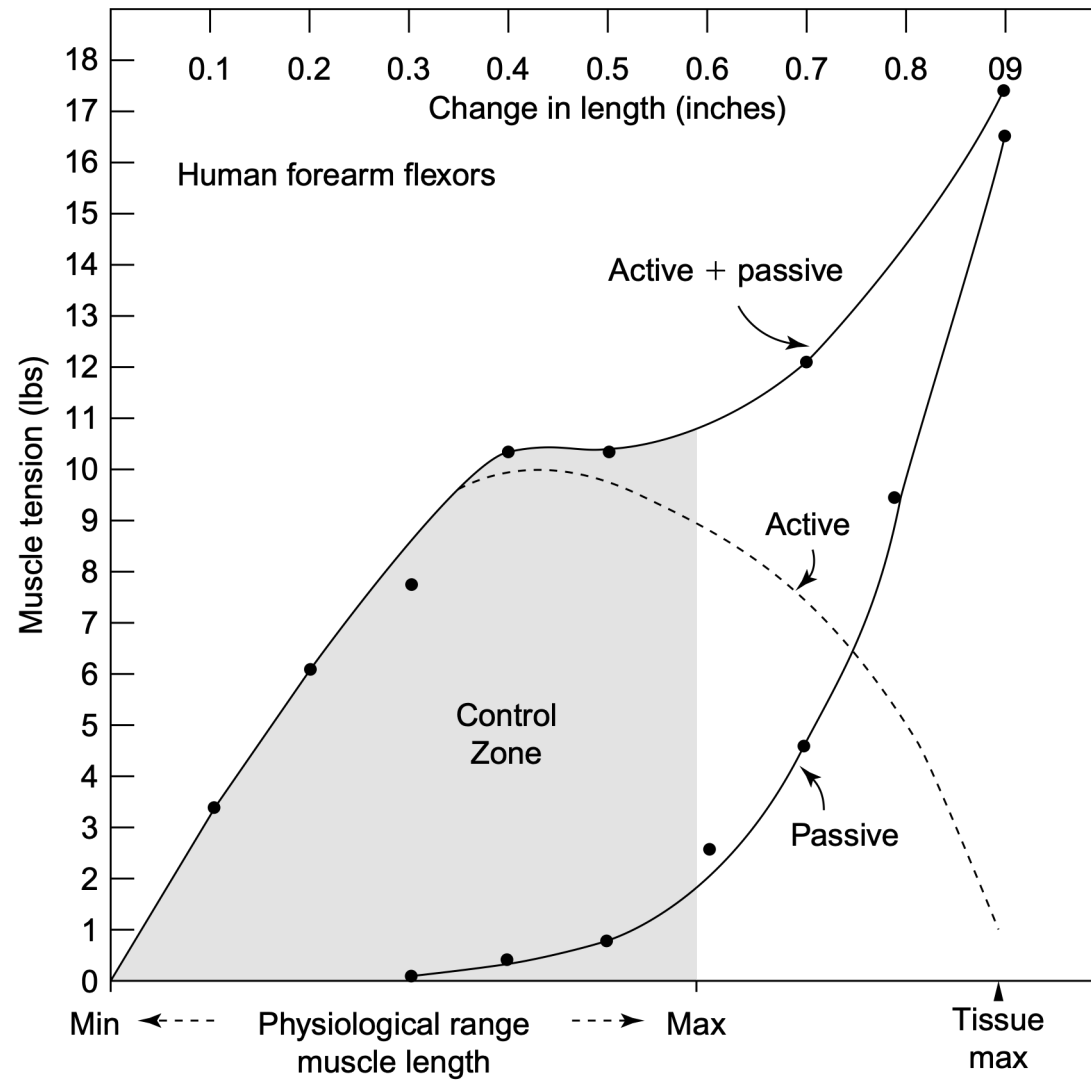
**Min.
overlap**



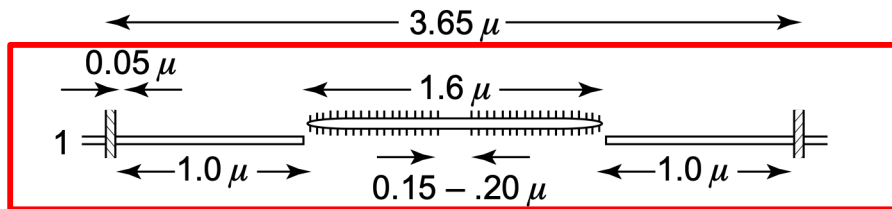
**Max.
overlap**

Energy required

Total tension

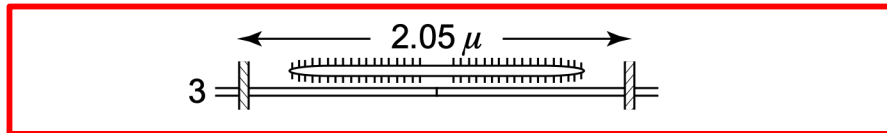
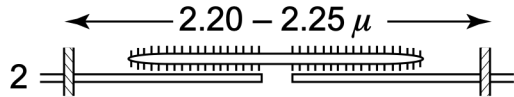


Total tension

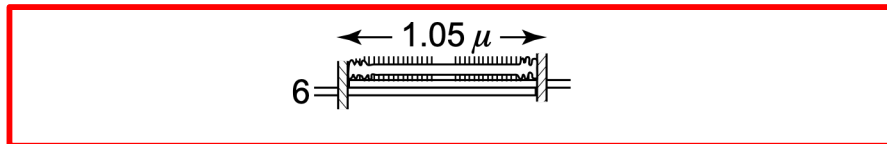
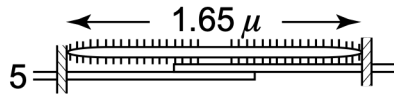
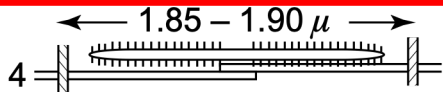


Min. overlap

Min Force



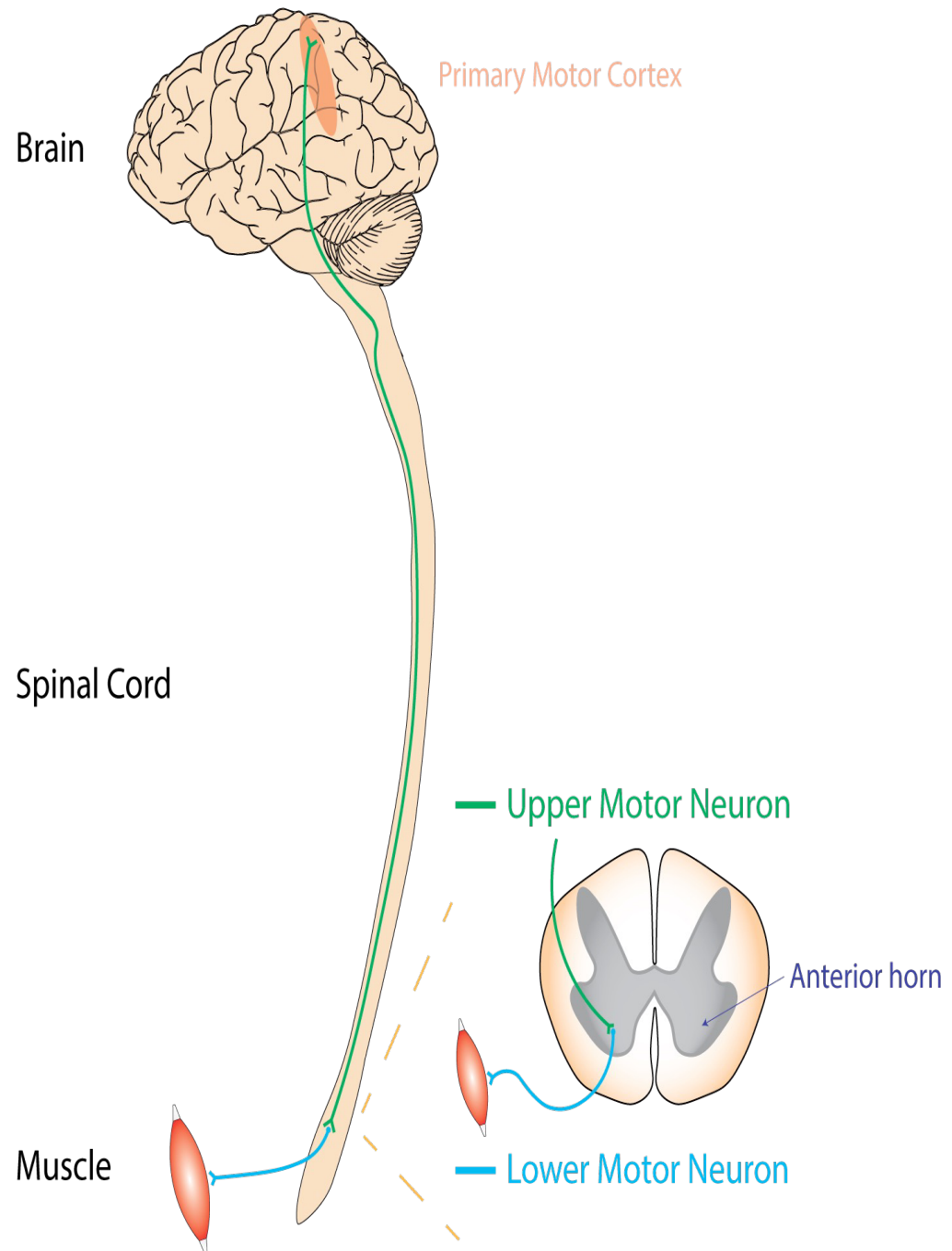
Max Force



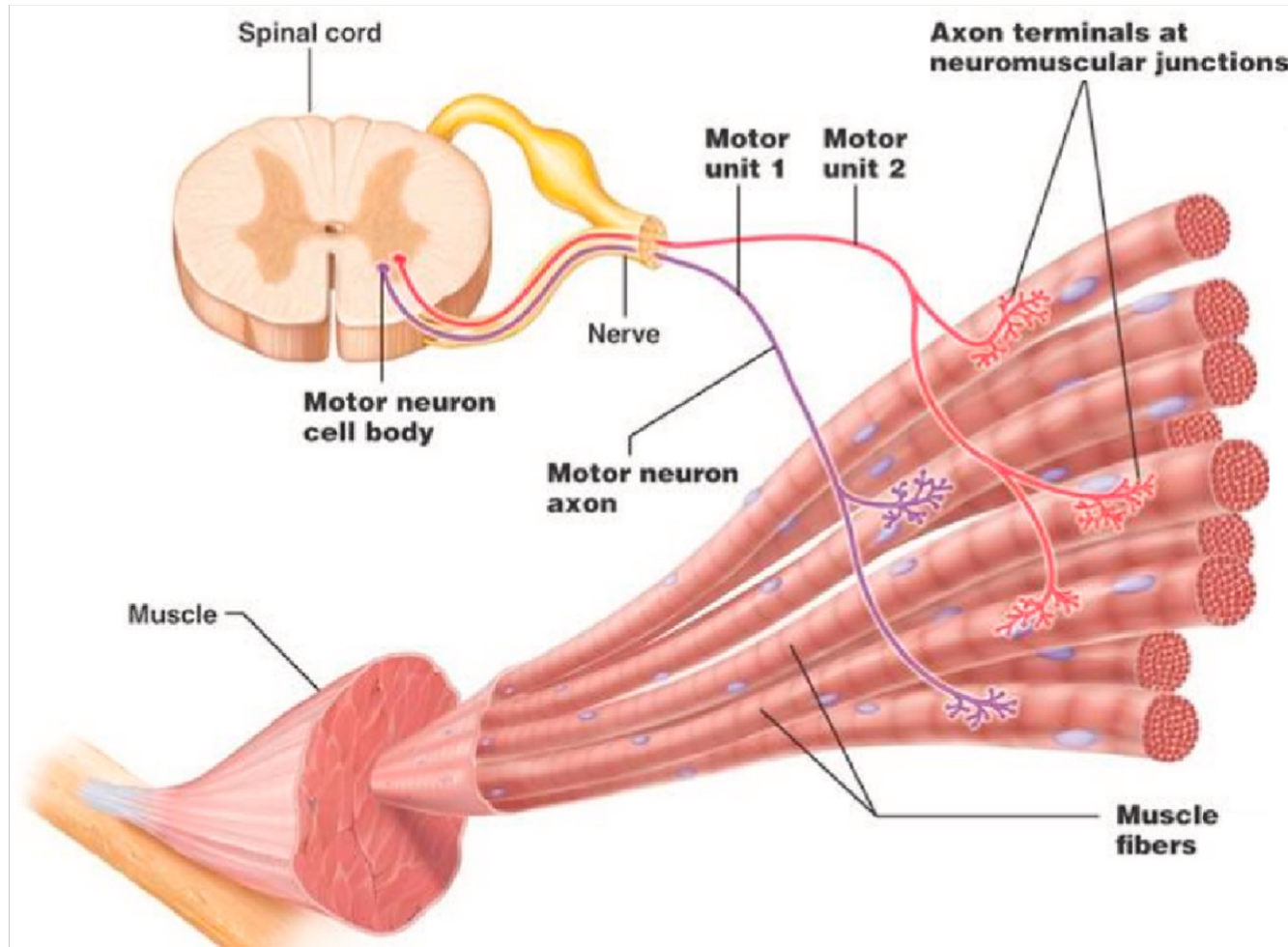
Max. overlap

Min Force

Motor Unit

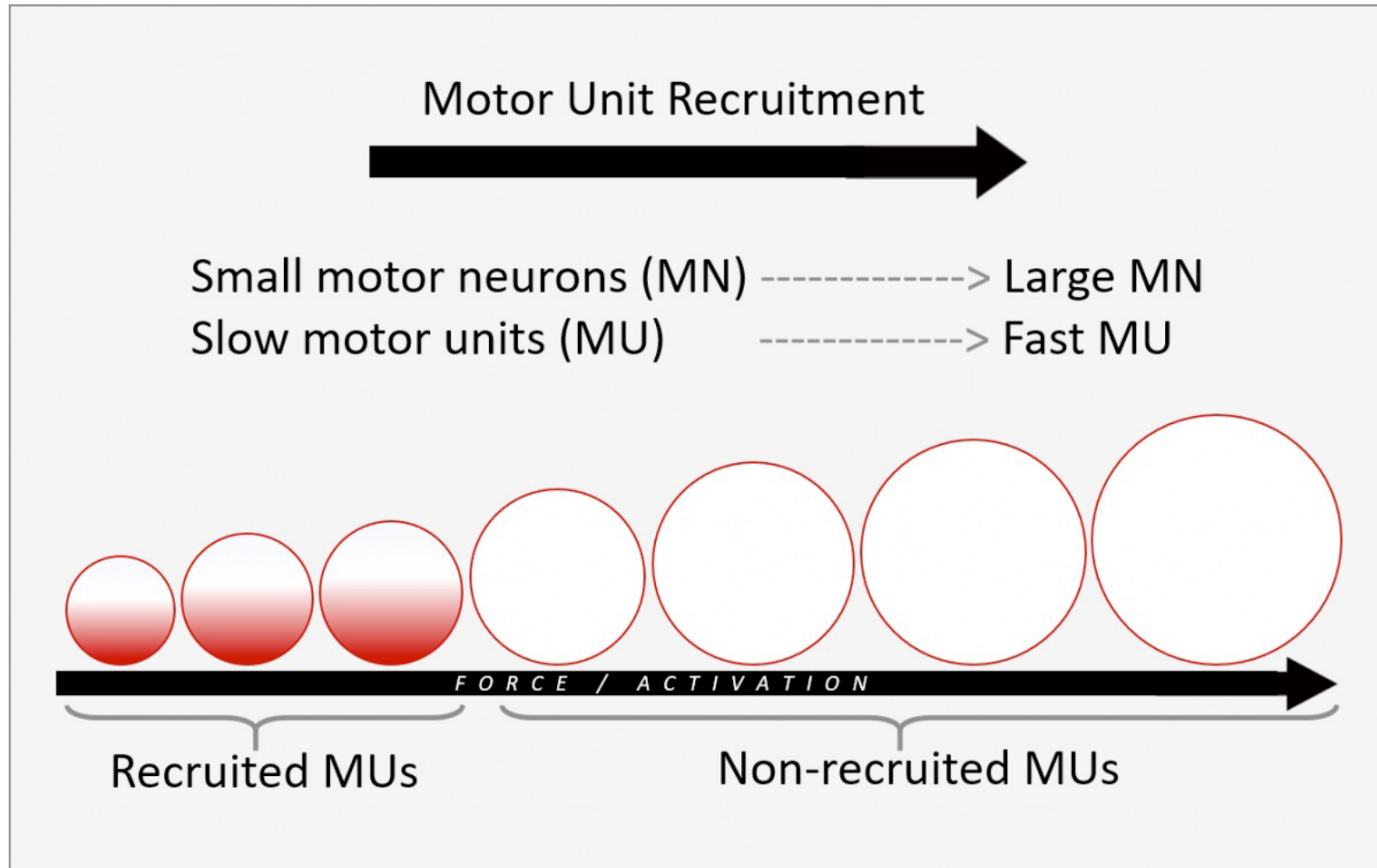


Motor Unit



Basic unit of Motor Control

Motor Unit



Size Principle